

In-Class Worksheet

Rates, Standardization, and Advanced R Programming

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Rates and Standardization, Advanced R Programming, AI Integration

1 Part 1: Review and Warm-up

1.1 Exercise 1.1: Quick Visualization Review

Using the guinea pig data, work with a partner to create one boxplot comparing all three groups:

```
1 # Recreate the data
2 control <- c(312, 345, 287, 398, 421, 356, 389, 334, 367, 401)
3 low_dose <- c(289, 267, 301, 234, 198, 245, 278, 213, 256, 298)
4 high_dose <- c(156, 189, 134, 167, 123, 145, 178, 201, 143, 134)
5
6 # Your code here:
```

Quick Questions:

- Which group has the most consistent survival times?
- How do the medians compare across groups?

2 Part 2: Introduction to Rates

2.1 Exercise 2.1: Understanding Rates in Public Health

Scenario: You're analyzing COVID-19 data from two counties:

County A: 150 cases out of 10,000 residents

County B: 300 cases out of 25,000 residents

Calculate and compare:

```
1 # County A
2 cases_A <- 150
3 population_A <- 10000
4 rate_A <- (cases_A / population_A) * 1000 # Rate per 1,000
5
6 # County B
7 cases_B <- 300
8 population_B <- 25000
9 rate_B <- (cases_B / population_B) * 1000 # Rate per 1,000
10
11 print(paste("County A rate per 1,000:", rate_A))
12 print(paste("County B rate per 1,000:", rate_B))
```

Fill in your results:

- County A raw cases: _____ out of _____
- County B raw cases: _____ out of _____

- County A rate per 1,000: _____
- County B rate per 1,000: _____
- Which county has the higher infection rate? _____

2.2 Exercise 2.2: Age-Standardized Rates

Extended Scenario: Now consider age breakdown:

County A:

- Ages 0-50: 80 cases out of 7,000 residents
- Ages 50+: 70 cases out of 3,000 residents

County B:

- Ages 0-50: 120 cases out of 15,000 residents
- Ages 50+: 180 cases out of 10,000 residents

```

1 # County A age-specific rates
2 county_A_young <- (80/7000) * 1000
3 county_A_old <- (70/3000) * 1000
4
5 # County B age-specific rates
6 county_B_young <- (120/15000) * 1000
7 county_B_old <- (180/10000) * 1000
8
9 print("Age-specific rates per 1,000:")
10 print(paste("County A (0-50):", round(county_A_young, 2)))
11 print(paste("County A (50+):", round(county_A_old, 2)))
12 print(paste("County B (0-50):", round(county_B_young, 2)))
13 print(paste("County B (50+):", round(county_B_old, 2)))

```

Analysis Questions:

1. Which age group has higher infection rates in both counties?
2. How does age structure affect the overall county comparison?
3. Why might age-standardization be important for fair comparisons?

3 Part 3: AI-Enhanced Rate Calculations

3.1 Exercise 3.1: Automating Rate Calculations with AI

AI Challenge: Use AI to help create a function that calculates and compares rates between populations.

Your AI Prompt: “Help me create an R function that calculates rates per specified population size (like per 1,000 or per 100,000) and compares multiple groups. Include error checking and clear output formatting.”

AI Response:

```

1 # Paste AI-generated function here:

```

Test the Function:

```

1 # Test with our county data
2 # Your test code here:

```

Verification Steps:

- ☐ Does the function calculate rates correctly?
- ☐ Does it handle different population bases (per 1,000, per 100,000)?
- ☐ Are there appropriate error checks?
- ☐ Is the output clear and interpretable?

3.2 Exercise 3.2: Advanced AI Prompt for Epidemiological Analysis

Advanced AI Prompt: “I need to perform a comprehensive epidemiological analysis comparing disease rates across different populations with age standardization. Create R code that handles multiple age groups, calculates both crude and age-adjusted rates, and provides appropriate visualizations.”

Document AI Response:

1. Key functions suggested:

2. New concepts learned:

3. Code that needed modification:

4 Part 4: Advanced R Programming

4.1 Exercise 4.1: Creating Functions

Based on our rate calculations, create a reusable function:

```
1 # Template function - complete the missing parts
2 calculate_rate <- function(cases, population, per = 1000) {
3   # Your code here
4   # 1. Calculate the rate
5   # 2. Add error checking for negative values
6   # 3. Return formatted result
7
8 }
9
10 # Test your function
11 calculate_rate(150, 10000, 1000)
12 calculate_rate(300, 25000, 1000)
```

4.2 Exercise 4.2: Working with Real Data Structures

```
1 # Create a more complex dataset
2 health_data <- data.frame(
3   county = c("A", "A", "B", "B"),
4   age_group = c("0-50", "50+", "0-50", "50+"),
5   cases = c(80, 70, 120, 180),
6   population = c(7000, 3000, 15000, 10000)
7 )
8
9 # Calculate rates for each row
10 health_data$rate_per_1000 <- (health_data$cases / health_data$population)
    * 1000
11
12 # Group calculations
13 library(dplyr) # Load dplyr for data manipulation
14
15 # Calculate summary statistics by county
16 county_summary <- health_data %>%
17   group_by(county) %>%
18   summarise(
19     total_cases = sum(cases),
20     total_population = sum(population),
21     crude_rate = (total_cases / total_population) * 1000
22   )
23
24 print(county_summary)
```

Interpretation Questions:

1. What is the crude rate for each county?
2. How do these compare to your earlier calculations?
3. What additional information would help interpret these rates?

5 Part 5: Statistical Thinking and Study Design

5.1 Exercise 5.1: Connecting Rates to Biostatistics

Case Study Discussion: A pharmaceutical company reports their new drug reduces heart attack rates from 4 per 1,000 patients to 2 per 1,000 patients.

Questions for Discussion:

1. What does this represent in terms of:
 - Absolute risk reduction?
 - Relative risk reduction?

```
1 # Calculate the reductions
2 baseline_rate <- 4/1000
3 treatment_rate <- 2/1000
4
5 absolute_reduction <- baseline_rate - treatment_rate
6 relative_reduction <- (baseline_rate - treatment_rate) / baseline_rate
7
8 print(paste("Absolute risk reduction:", absolute_reduction))
9 print(paste("Relative risk reduction:", relative_reduction))
```

2. If you had 10,000 patients, how many heart attacks would be prevented?
3. What additional information would you want before recommending this drug?

5.2 Exercise 5.2: Bias and Confounding Preview

Looking back at our county COVID comparison:

1. What factors besides county might affect infection rates?
2. How might differences in testing rates affect our rate calculations?
3. What is the population we can generalize these results to?

6 Part 6: AI Integration and Documentation Practice

6.1 Exercise 6.1: Comprehensive AI Workflow

Task: Use AI to help analyze a complete epidemiological dataset and document your entire workflow.

Your Comprehensive AI Prompt: “I have epidemiological data with counties, age groups, cases, and populations. Help me create a complete analysis workflow that includes: 1) data validation, 2) crude rate calculations, 3) age-standardized rates, 4) appropriate visualizations, 5) statistical interpretation, and 6) recommendations for further analysis.”

Document Your AI Interaction:

1. Initial AI suggestions:

2. Code you tested:

```
1 # Paste and test AI code here:
```

3. Modifications needed:

4. Final verified analysis:

5. What you learned about AI capabilities and limitations:

7 Part 7: Preparation for Future Topics

7.1 Exercise 7.1: Probability Preview

When we cover probability theory, Based on today's work:

1. If County A's infection rate is 15 per 1,000, what's the probability that a randomly selected resident is infected?
2. How does this connect to what we've learned about populations and samples?

Bridge to Probability:

- Rates can be thought of as probabilities
- Population parameters vs. sample statistics
- Uncertainty in our estimates

8 Wrap-up Discussion

Key Concepts from Today:

- Rates allow fair comparison between different-sized populations
- Age standardization helps control for demographic differences
- Functions in R make calculations reusable and less error-prone
- AI can accelerate complex analyses but requires careful verification

Next Class Preview: We'll begin with probability concepts - the foundation for all statistical inference.

9 AI Usage Documentation

Complete Documentation Required:

Tools Used: _____

Key Prompts:

1. _____
2. _____
3. _____

AI-Generated Code Verification:

- Code tested and working: ☐ Yes ☐ No
- Modifications made: _____
- Errors found: _____

Learning Assessment:

- Most useful AI contribution: _____
- Biggest AI limitation encountered: _____
- Confidence in AI-assisted results (1-10): _____
- Independent verification methods used: _____

Next Steps for AI Integration:
